

Order Flow and Cryptocurrency Returns: A Machine Learning Approach with Out-of-Sample Validation

Abstract

The results indicate that overall order flow plays a significant role in explaining variations in cryptocurrency returns. Using machine learning for out-of-sample predictions, we demonstrate that order flow is an effective predictor of daily cryptocurrency returns. Portfolio strategies that rely on machine-learning forecasts based on daily order flow achieve an alpha of up to 0.86% per day and a Sharpe ratio of 3.97 annually. Order flow outperforms economic fundamentals in predicting returns, especially when nonlinear machine learning models conditioned on order flow are used, demonstrating strong out-of-sample predictive power beyond arbitrage limits. In addition to statistical analysis, we assess the economic value of conditioning on order flow through portfolio sorting. Overall, our findings suggest that order flow consistently influences cryptocurrency returns.

Keywords: Order flows, cryptocurrency returns, machine learning techniques, and out-of-sample.

1. Introduction

The cryptocurrency market has experienced rapid expansion over the past decade, with market capitalisation rising from \$428.60 billion in 2018 to \$ 3,257.43 trillion in 2025, and daily trading volume increasing from \$160 million to \$415 billion USD in 2025. Unlike traditional foreign exchange markets, which are dominated by a few major dealers [22], the crypto market features independently operated, non-integrated exchanges that operate across various countries and currencies [21]. Despite its growth, the market remains illiquid and faces considerable information asymmetry compared to conventional financial markets [3]. These unique qualities position crypto markets as a fresh domain for studying how order flow influences price formation.

Order flow indicates the total buy and sell volumes and greatly influences price, liquidity, and volatility. [6] highlight order flow as the immediate factor affecting price in almost all microstructure analyses, underscoring its importance for understanding short-term market behaviour. It measures the net load of a cryptocurrency by calculating the natural-log difference between buyer- and seller-initiated transaction volumes, denominated in the relevant fiat currency over a specified period. The link between order flow and cryptocurrency returns is driven primarily by economic indicators rather than by order flow itself, supporting the “no-predictability view” and the null hypothesis we examine. Since order flow is believed to have only a temporary effect on crypto returns, it is seen as a transitory phenomenon, a view known as the “transitory view” [12, 22]. In contrast, order flow also reflects the integration of future economic information into cryptocurrency returns through price discovery, often referred to as the “permanent view” [6, 7, 22].

The analysis of order flow also examines how various events in the order book influence future price changes. Modelling order flow and its market impact is crucial in finance for understanding how private information is reflected in prices and for creating profitable trading strategies. In cryptocurrency markets, order flow has a temporary effect on returns, but this influence doesn't last long. Microstructure research examines how these short-term effects

relate to shifts in liquidity, inflationary trends, and short-term shocks to preferences. In-sample data show that global order flows are the strongest predictor of cross-sectional crypto returns, with a positive correlation. These flows, along with control variables, explain about 15-22% of the variation in returns. The regression emphasises the importance of distinguishing between short-lived, transient order-flow components and longer-term, persistent ones. Following [3], lagged returns are used as proxies for short-term reversals. Analyses with lagged order flow and past crypto returns reveal that a significant portion of lagged flow, even when uncorrelated with past returns, still strongly predicts future returns. The positive relationship between global order flow and daily or weekly crypto returns supports the idea that order-flow effects are persistent. Panel regressions show that the most influential coefficient among all global order flows is from these flows. When global order flow is included in regressions, U.S. order flow remains highly significant, though its effect diminishes. Overall, these results suggest that global order flows are more active and are better predictors than individual flow components. Out-of-sample forecasting of typical cryptocurrency returns using order flow employs machine learning (ML) techniques for variable selection and dimensionality reduction, accommodating a large number of indicators and many usable forms. Machine learning-based forecasting allows us to incorporate global order flows and other economic indicators to predict the next day's cryptocurrency returns.

The estimation of the linear and the nonlinear ML techniques, including ridge regression (RR), Lasso (AS), elastic net (EN), principal component regression (PCR), and random forest (RF), stochastic gradient boosted regression trees (SGB), as well as the neural networks comprising 1-4 hidden layers (NN1-NN4) along with the varied forest combinations spanning several linear models (L-Avg) and also the non-linear models (NL-Avg), as the out-of-sample ML forecasting is only carried out for the regular returns. Regarding out-of-sample performance, the non-linear ML forecasts, conditioned on all order flows, consistently surpass the linear models conditioned on all order flows, as well as the linear and non-linear models based on the economic indicators and the zero-forecast benchmark. From amongst these, the standard gradient boosted regression model (SGB) that conditions on almost all the order flows with R^2 of 0.41%. The non-linear ML techniques that condition on order flow display significant out-of-sample predictive ability.

2. Research Literature

This study advances the growing cryptocurrency literature by analysing the information entropy of order flow to assess and predict cryptocurrency returns. We examine three main questions: whether order flow can explain cross-sectional cryptocurrency returns, if it provides predictive insights, and whether key information in international order flows is missing from US order flow data. Addressing these questions is essential to understanding market microstructure and supports broader research into the economic factors that shape order flow and price discovery in cryptocurrency markets. While these questions have been answered in traditional financial markets, they remain open in the cryptocurrency context. By replicating the information contained in order flow within the cryptocurrency market, this paper fills a gap in existing research. Several studies have shown empirical evidence of order flow's impact on physical currency exchange rates. We primarily focus on the out-of-sample test, which provides the most in-depth evidence about the asset return predictability [27, 29]. Very few studies analyse out-of-sample cryptocurrency return prediction; for example, [7,6,15] use various technical indicators to predict daily and weekly Bitcoin returns, whereas others predict

daily Bitcoin returns using several predictors. We also extend the existing cryptocurrency returns literature by exploring the out-of-sample predictability of daily cryptocurrency returns, applying a broad range of predictors and a wide variety of machine learning techniques. The daily cryptocurrency return forecasts are generated using various machine-learning methods, including elastic net [32] and linear model estimation that includes almost all indicators (Linear-ENet forecast). Conventional Ordinary Least Squares (C-OLS) estimates a large-scale linear model using correlated economic indicators, leading to in-sample overfitting. The ENet refers to an extension of the seminal least absolute shrinkage and selection operator [29], helping to mitigate overfitting by using penalised (Lasso) regression to shrink coefficient estimates to near zero. [6, 7] develop a theoretical model of order flow, confirming that it significantly influences the Deutschmark/dollar market, where order flow explains at least 50% of exchange-rate variation and yields residuals that outperform the random-walk benchmark. [2] Use high-frequency intraday data to find a strong positive relationship between order flows and crypto returns across time horizons ranging from 1 minute to 1 day. [22] prioritise fiat currencies and evaluate the breakdown of foreign exchange customer segments by order flow, suggesting that order flow contains important forecast information, though the degree of uniformity varies across end-user foreign exchange markets. The literature on market microstructure links transitory market effects to variations in liquidity, price impact, and temporary preference shocks [12, 22]. Several studies based on market microstructure correlate permanent effects with asymmetric information among market participants, such that transactions convey information and have a consistent effect on security prices [13, 18, 15, & 16]. Crypto trades convey information about future economic indicators that is unknown to all market participants, and order flow then serves as the primary vehicle for spreading this information about cryptocurrency returns through the price discovery process [6, 7, 22]. Several papers assess the out-of-sample predictive power of order flow for daily crypto returns using machine learning (ML) techniques [14, 6, 10, 13]. [2] Analyse the profitability of machine learning models under economic constraints, such as short-selling limitations, transaction costs, and limits on speculators' risk-taking capacity.

Hypothesis

The hypotheses analysed in the study are

1. The order flow $OrFlow_k$ positively affects the linear relationship with simultaneous mid-level price change $\Delta MidPr_k$, i.e., the price change in the coefficient β_{OrFlow_k} is not equal to zero and is statistically significant at 1% significance level, i.e., $p \leq 0.01$.

$$H_0: \beta_{OrFlow} = 0;$$

$$H_1: \beta_{OrFlow} \neq 0;$$

2. The international trade flows balance ITF_k positively affects the linear correlation with the simultaneous mid-level price change $\Delta MidPr_k$ i.e., the price change in the coefficient β_{ITF_k} is not equal to zero and is statistically significant at 1% significance level, $p \leq 0.01$.

$$H_0: \beta_{ITF} = 0;$$

$$H_1: \beta_{ITF} \neq 0;$$

3. The order flow $OrFlow_k$ firmly explained the international trade flow imbalances ITF_k on the mid-level price variation $\Delta MidPr_k$, similarly as computed by the coefficient of determination- R-squared (R^2)

3. Data and Methodology

We use cross-sectional data on 128 cryptocurrencies for the sample period from February 14th, 2020, to June 30th, 2025. For each cryptocurrency or coin, the daily price data is sourced from coinmarketcap.com (CMC) at 00:00 am GMT. The data collected from CMC comes from over 700 exchanges and is therefore considered a reliable source, frequently used in the crypto literature, such as [24] and [25], which exclude weekends and holidays from the daily sample. CMC contains data on thousands of coins with low market capitalisation and limited or historical data; hence, to obtain the final cross-sectional data for 128 coins, we apply two criteria to ensure liquidity across the entire dataset. The first criterion is that the coins must have a market capitalisation greater than 1 million USD on the first day of the sample period. The second is that the coins must be traded continuously at non-zero prices and volumes in each period of the sample. As a result, our dataset comprises a balanced panel of 128 coins listed in Appendix Table 1. We use the daily prices to compute their respective returns.

$$re_{i,t} = \frac{Pr_{i,t}}{Pr_{i,t-1}} - 1 \quad (1)$$

where, $Pr_{i,t}$ refers to the price of coin i at time t , and the weekly crypto returns are calculated over the period from Saturday at 00:00 GMT to the following Friday at 23:59 GMT, covering seven days including the weekend. at 11.59 pm GMT, spanning seven days, including the weekend.

The volatility is assessed, like the five-minute annualised log return volatility.

$$\sigma = \sqrt{Var\left(N \log\left(\frac{Pr_t}{Pr_{t-1}}\right)\right) * 288 * 365} \quad (1.1)$$

3.1 Methodology

Machine learning method

We index coins as $I = 1, \dots, N$ and days $t = 1, \dots, T$, and describe the returns of the coins I at time $t + 1$ as

$$re_{i,t+1} = E_t(re_{i,t+1}) + \epsilon_{i,t+1}, \quad (2)$$

where, $E_t(re_{i,t+1}) = g(z_{i,t})$ refers to the conditional expected return, and $g(\cdot)$ displays as the flexible function of the coin i 's P -dimensional vector of characteristics $z_{i,t}$, aided by Gu et al. (2020), we estimate $g(\cdot)$ making use of a variety of models with increasing complexity to maximise the out-of-sample predictive power for the realised $re_{i,t+1}$. In asset return prediction using machine learning techniques, such as those of [15] and [19], among others, the pooled approach is used to forecast individual stock returns. [12] Also, use the above approach to forecast exchange rate changes. The excess return forecast is given by:

$$\widehat{re}_{i,t+1} = \hat{f}(x_{i,t}; \hat{\eta}) \quad (3)$$

where, $\hat{f}(\cdot; \hat{\eta})$ refers to the fitted predictive model based on data through day t , ensuring no forward-looking bias in the forecast.

For each Monte Carlo sample, we divide the full time series into 3 sequential subsamples of equal length for training, validation, and analysis. Especially, we evaluate each of the two models included in the training model by applying PLS, PCR, Ridge, Lasso, Elastic Net (ENet), a generalised linear model with the Lasso group (GLM), random forest (RF), gradient boosted regression trees (GBRT), and the five similar architecture frameworks of the neural networks (NN1,..., NN5) we approve the empirical analysis, respectively.

Linear-ENet

A linear specification for the prediction model is considered, and we fit it using ENet [35]. The linear ENet excess return forecast is summarised from the intercept term for minimalism, which is expressed as:

$$\widehat{re}_{i,t+1}^{ENet} = x'_{i,t} \hat{\eta}, \quad (4)$$

where, $\hat{\eta}$ refers to the vector of coefficients for the linear model estimated via ENet using data up to time t . Conventional OLS estimation of the linear model maximises the fit of the model over the training model. The ENet is often regarded as an extension of the original LASSO [32], in which, like LASSO, it uses penalised regularised regression to shrink parameter estimates towards zero, thereby guarding against overfitting. The ENet objective function applied to the linear model is expressed as:

$$\arg \min_{\eta} \frac{1}{2(t-1)n} \left[\sum_{i=1}^n \sum_{s=1}^{t-1} (re_{i,s+1} - x'_{i,s} \eta)^2 \right] + \lambda P_{\delta}(\eta), \quad (5)$$

Where, $P_{\delta}(\eta) = 0.5(1 - \delta) \|\eta\|_2^2 + \delta \|\eta\|_1$; and $\lambda \geq 0$ refers to the hyperparameter that directs the degree of shrinkage; $\|\cdot\|_1$ and $\|\cdot\|_2$ stands for the $loss_1$ and $loss_2$ standards, respectively; $0 \leq \delta \leq 1$ refers to the hyperparameter for mixing the $loss_1$ and $loss_2$ elements of the penalty term; and n is the number of cryptocurrencies. For the ease of the individual, the equation above uses a balanced panel. For the analysis in our paper, as it is very simple to change the notation for the unbalanced panel. The ENet objective function shown in the above equation reduces the values for the OLS parameter when $\lambda = 0$ and if the value is $\delta = 1$, the resultant value equals the LASSO objective function, whereas the $\delta = 0$ presents the ridge regression [22]. It also encompasses an $Loss_1$. Components like the lasso objective function, as it permits shrinkage to precisely zero, and ENet also performs variable selection.

Order Flow

Order flow measures the log difference between customer-initiated and seller-initiated transaction volumes in a specific fiat currency over time. We compute order flow using signed volume data (buy and sell volumes) from cryptocompare.com (CC), a reliable source of cryptocurrency volume data [4]. CC compiles data from over 300 exchanges and provides signed volume information that coinmarketcap.com (CMC) does not offer. In our study, we use crypto data from CMC, except for signed volume or order flow, which is available only from cryptocompare.com. The advantage of CC data is that it supplies signed volume figures in multiple fiat currencies. We examine order flow across the G10 currencies, as well as in South Korea, an important trading hub outside the G10. This group is called the G11 currencies,

including the US dollar (USD), Euro (EUR), British pound (GBP), Japanese yen (JPY), Swiss franc (CHF), Canadian dollar (CAD), Australian dollar (AUD), New Zealand dollar (NZD), Norwegian krone (NOK), Swedish krona (SEK), and Korean won (KRW). We present a measure of order flow comparable across cryptocurrencies. Following [28], we standardise each series as follows:

$$OrFlow_{i,t} = \frac{of_{i,t}}{\sigma(of_{i,t-29:t})} \quad (5)$$

where $OF_{i,t}$ shows the standardised order flow of coin i at time t , $of_{i,t}$ refers to the original order flow, and $\sigma(of_{i,t-29:t})$ shows order-flow volatility over the last 30 days. Due to standardisation, the first day of our sample is February 14th, 2018, as the standardised order flow is shortened to address potential large volatility spikes; hence, the regression results did not change significantly when using truncated vs. non-truncated order flow. Additionally, to the 11 international order flows, we also determine the cumulative measure known as the world order flow ($OrFlow^{WF}$), whereby, it is calculated by aggregating the 11 buy and sell volumes, taking the log difference, and applying the same standardisation as for the disaggregated order flows. To verify the contribution of each cryptocurrency to the world order flow, we conduct a variance decomposition as follows:

$$VarCon_i^{WF} = \frac{\beta_i^2 Var(OrFlow_i)}{\sum_{j=1}^{11} \beta_j^2 Var(OrFlow_j)} \quad (6)$$

where $VarCon_i^{WF}$ is the variance contribution of coin i to the world order flow, β_i displays the estimated coefficient of each order flow when regressing the world order flow on its 11 components, and $Var(OrFlow_i)$ refers to the variance of each order flow.

Linear Regression

The benchmark model in our paper does not include nonlinearities or interactions among predictors. It assumes that the conditional expectation of return $g(\cdot)$ can be approximated by a linear function of the predictors and the parameter vector θ .

$$g(z_{i,t}; \theta) = z'_{i,t} \theta \quad (7)$$

The pooled ordinary least squares (OLS) are obtained by minimising the l_2 objective function:

$$\mathcal{L}(\theta) = \frac{1}{No.ofObs.T} \sum_{i=1}^N \sum_{t=1}^{obsT} (re_{i,t+1} - g(z_{i,t}; \theta))^2 \quad (8)$$

In our paper, the OLS estimator remains unbiased and fairly efficient when the number of predictors, P , is small relative to the number of observations, T . However, as P approaches T , the estimator becomes inefficient and prone to overfitting. To mitigate this problem, we utilise several Machine Learning (ML) models that have recently been applied in the asset pricing literature.

Birched Linear Regression: Elastic, Lasso, and Ridge Net

The Accelerated Proximal Algorithm (APG) of [32] and [34], which allows for the efficient implementation of the elastic net, Lasso, ridge regression and Group Lasso for both $loss_2$ and

Huber losses. This method reduces overfitting in OLS by shrinking the least-squares estimates toward zero by adding a penalty term to the objective function. This standard procedure intentionally deteriorates the model's in-sample fit to enhance its out-of-sample stability. Birch linear models estimate θ by reducing:

$$\mathcal{L}(\theta; \cdot) = \mathcal{L}(\theta) + \phi(\theta; \cdot) \quad (9)$$

Where, $\mathcal{L}(\theta)$ refers to the loss function and the $\phi(\theta; \cdot)$ shows the penalty ad and also omits the dependency on the tuning parameters. Specifically, we calculate

$$Ridge = \frac{1}{2} \lambda \sum_{j=1}^{Pr} \theta_j^2, Lasso = \lambda \sum_{j=1}^{Pr} |\theta_j| \quad (10)$$

$$Elastic Net = \lambda(1 - \rho) \sum_{j=1}^{Pr} |\theta_j| + \frac{1}{2} \lambda \rho \sum_{j=1}^{Pr} \theta_j^2, Group Lasso = \sum_{j=1}^{Pr} \|\theta_j\| \quad (11)$$

Where, in the Lasso Group case, $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ is the $K \times P$ matrix and where $\phi(\theta; \cdot)$ is the commonly used elastic net (EN) penalty on θ defined as:

$$\phi(\theta; \lambda, \rho) = \lambda(1 - \rho) \sum_{j=1}^P |\theta_j| + \frac{1}{2} \lambda \rho \sum_{j=1}^P \theta_j^2 \quad (12)$$

where λ refers to the positive tuning parameter controlling the degree of the shrinkage, so that the enhanced values of λ lead to further shrinkage of the estimated parameters. When $\lambda=0$, the model reduces to OLS, and when $\lambda > 0$, the EN incorporates two famous standard procedures as special cases, controlled by the positive hyperparameter ρ . When $\rho = 0$, equation (1) equals the Lasso (LAS), which enforces an l_1 parameter penalisation. LAS may specify a subset of θ to exactly zero, imposing rarity on the model and thus performing variable selection. When $\rho = 1$, equation (2) equals the ridge regression (RR), which enforces an l_2 parameter penalization. Ridge regression decreases the coefficient estimates toward zero while keeping all predictors in the model. For all the values of ρ between 0 and 1, the elastic net blends the benefits of RR and LASSO by performing both shrinkage and variable selection. We enhance the hyperparameters, λ and ρ , by using the validation sample.

Proximal Algorithms

Proximal algorithms are used to solve convex hyperbolic optimisation problems, where the underlying operation is to evaluate the function's recursive operator, such as solving the optimisation problem, and, in most cases, the smaller problem has a closed-form solution. The optimisation problem requires the proximal operators of $\phi(\theta; \cdot)$ s which have a closed form

$$\begin{aligned} ProximAlgo_{\gamma\phi}(\theta) &= \{ Ridge = \frac{\theta}{1 + \lambda\gamma} \\ &\quad \{ Lasso = \lambda S(\theta, \lambda\gamma) \\ &\quad \{ Elastic Net = \frac{\theta}{1 + \lambda\gamma\rho} S(\theta, (1 - \rho)\lambda\gamma), \\ &\quad \{ Group lasso \\ &= (\tilde{S}(\theta_1, \lambda\gamma)^T, \tilde{S}(\theta_2, \lambda\gamma)^T, \dots, \tilde{S}(\theta_p, \lambda\gamma)^T)^T \end{aligned} \quad (13)$$

where, $S(x, \mu)$ and $\tilde{S}(x, \mu)$ are the multivariate functions where the i th Components are defined by

$$(S(x, \mu))_i = \begin{cases} x_i - \mu & \text{where if } x_i > 0 \text{ and } \mu < |x_i|; \\ x_i + \mu & \text{where if } x_i < 0 \text{ and } \mu < |x_i|; \\ 0, & \text{if } \mu \geq |x_i|; \end{cases} \quad (14)$$

$$(\tilde{S}(x, \mu))_i = \begin{cases} x_i - \mu \frac{x_i}{\|x_i\|}, & \text{if } \|x_i\| > \mu; \\ 0, & \text{if } \|x_i\| \leq \mu. \end{cases} \quad (15)$$

Here, the $S(x, \mu)$ refers to the binarisation function so that the proximal algorithm is equal to the coordinate descent algorithm in the case of loss l_2 like [8], [16].

Principal Component Analysis Regression (PCAR)

The PCAR technique is a dimension-reduction method that uses a two-step procedure. In the first stage, principal component analysis (PCA) compresses the P predictors into K linear combinations that best capture their common variation. In particular, PCA linearly transforms the set of predictors into orthogonal principal components, with the initial component having the highest variance, and the second component having the second highest variance, and so on, up to where the condition $K < P$ principal components. The j^{th} principal component Zw_j is used to solve the optimisation problem:

$$w_j = \arg \max_w Var(Zw), \quad s.t. w'w = 1, \quad Cov(Zw, Zw_l) = 0, \quad l = 1, 2, \dots, j-1 \quad (16)$$

where Z stands for $NT \times P$ matrix of the basic standardised predictors, $z_{i,t}$ and w_j is the eigenvector corresponding to the j th eigenvalue of $Var(Z)$. In the second step, PCAR uses the first K leading components as regressors in the OLS regression. Hence, PCAR acts as a regularisation technique by isolating coefficients with very low variance components. The equation above shows that the first K principal components are selected to explain the variation in the P predictors. Though forecasting the future, there is no guarantee that these principal components are effective since $re_{i,t+1}$ is not considered in the above equation. Therefore, the potential drawback of PCAR is that the initial K principal components, while displaying high variance, may have little relation with $re_{i,t+1}$.

Random Forest and gradient boosted regression trees

Regression trees are fully non-parametric and use a different logic from linear models. A tree is built through a sequence of actions, where at each step a new “branch” splits the remaining data from the preceding action based on a single predictor. This sequential process divides the predictor area into rectangular regions, and the final result is the average values of $re_{i,t+1}$ within each region. In particular, the prediction of a tree with K regions and depth L , is calculated as:

$$g(z_{i,t}; \theta, K, L) = \sum_{k=1}^K \theta_k 1_{(z_{i,t} \in C_k(L))} \quad (17)$$

where, $C_k(L)$ refers to the K regions into which the data is partitioned, $1_{(\cdot)}$ is the indicator function depicting whether $z_{i,t}$ falls into the region where $C_k(L)$ and θ_k are a constant. We use the algorithm of [5] for the estimation of θ . For each new level, the predictor is selected, and the trigger value that minimises the prediction error is used to create the resulting two branches. This technique, known as recursive binary splitting, is considered a greedy algorithm because at each step of the tree-building process, the best split is selected based solely on that step, without considering future splits. The prediction error for each branch C is calculated as:

$$H(\theta, C) = \frac{1}{|C|} \sum_{z_{i,t} \in C} (re_{i,t+1} - \theta)^2 \quad (18)$$

here $|C|$ shows the number of observations within the particular branch C , where the loss function naturally leads to the optimal choice of θ as the test average of return within each branch, i.e., $\theta = \frac{1}{|C|} \sum_{z_{i,t} \in C} (re_{i,t+1})$. The branching continues until the maximum specified depth L is attained or the number of observations in each region is below the specified count. The trees capture complex non-linearities and adaptable interactions among the predictors, yet this flexibility also introduces a limitation that often leads to trees overfitting the training sample. To tackle overfitting, we employ two types of ensemble models that improve stability by averaging across multiple trees. The initial model is the random forest (RF), a special case of the general technique known as bagging. The partitions are generated by a sequence of splitting rules, generally based on the classification and regression tree (CART) algorithm. In regression, the predicted value is the mean of the target observations in the terminal nodes. The forecasts related to the regression tree with U terminal nodes is expressed as:

$$\widehat{re}_{i,t+1}^{RgressTree} = \sum_{u=1}^U \bar{re}_u 1_u(x_{i,t}; \hat{\eta}_u), \quad (19)$$

where the characteristic function $1_u(x_{i,t}; \hat{\eta}_u) = 1$ if $x_{i,t} \in R_u(\hat{\eta}_u)$ for the u th region denoted by R_u Correlated with vector $\hat{\eta}_u$ based on the splits characterising the tree, or else zero and $\bar{re}_u$ shows the mean value of the target sample data observations where Re_u stands for the training sample.

Forecast combinations

Linear and non-linear machine learning techniques help form forecast combinations by aggregating predictive ability across multiple models [39]. Following [37], we can compute the weighted average of all forecasts across the set of models at each time point, known as the mean combination. Conventionally, if the forecast of a given model j ($j = 1, \dots, J$) is denoted by $\widehat{re}_{i,t+1}^{(j)}$, then the forecast combination at $t + 1$ is:

$$\widehat{re}_{i,t+1}^{Mean} = \frac{1}{J} \sum_{j=1}^J \widehat{re}_{i,t+1}^{(j)} \quad (20)$$

In this context, J denotes the set of combined models. We examine and compare forecast combinations in two scenarios: (1) linear models (L-mean) and (2) non-linear models (NL-mean). This comparison allows us to evaluate the improvement in predictive accuracy achieved by integrating non-linear components and interactions among predictors.

Robust Huber Objective Function

Cryptocurrency returns follow a heavy-tailed distribution due to asymmetry and high kurtosis. The l_2 objective function shown in equation... reveals significant outliers that can destabilise the models. To mitigate the influence of this fat-tailed data and enhance model stability, we employed the robust Huber objective function, as supported by [21], which is illustrated as:

$$\mathcal{L}_H(\theta) = \frac{1}{No.ofObs.T} \sum_{i=1}^N \sum_{t=1}^{obsT} H(re_{i,t+1} - g(z_{i,t}; \theta), \xi), \quad (21)$$

$$\text{Here, } H(x; \xi) = \begin{cases} x^2 & \text{if } |x| \leq \xi; \\ 2\xi|x| - \xi^2 & \text{if } |x| > \xi \end{cases}$$

The Huber Loss $H(\cdot)$ integrates the L_2 loss reflecting the small errors and the l_1 loss, depicting the large errors. The threshold parameter ξ regulates the mix of l_1 and l_2 . By setting the threshold at the 99.9th percentile of returns across the joint training and validation data, we can efficiently handle extreme values. We train all our models, excluding Random Forest, using the Huber objective function, which is inherently robust to outliers, by following [21], and thereby use the default l_2 loss function. Estimation models that use the Huber objective function generally outperform those estimated with the mean squared error (MSE) objective function on cryptocurrency data.

Deep Neural Network- Feed-Forward (FF) Neural Networks

Neural networks are highly adaptable and can approximate any continuous function, similar to the human brain [25]. We utilise feed-forward neural networks with three layers: the input layer, which accepts raw predictors; one or more hidden layers that process and transform these predictors; and the output layer, which combines the hidden layers' outputs to produce the final projection. Each layer takes the signals from the neurons in the previous layer to generate the new signal:

$$Sh_m^{(l)} = g \left(wt_{m,0}^{(l)} + \sum_{j=1}^{P_{t-1}} wt_{m,j}^{(l)} h_j^{(l-1)} \right) \text{ for } m = 1, \dots, P_t; l = 1, \dots, L \quad (22)$$

Here, $Sh_m^{(l)}$ refer to the signal related to the m th neuron in the l th hidden layer; $wt_{m,0}^{(l)}, wt_{m,1}^{(l)}, \dots, wt_{m,P_{t-1}}^{(l)}$ are the corresponding weights; and $g(\cdot)$ shows the nonlinear activation function. The final layer known as the output layer that derives the signal from the last hidden layer and finally transforming them into the prediction:

$$\widehat{re}_{i,t+1}^{Net} = wt_0^{(L+1)} + \sum_{j=1}^{P_L} wt_j^{(L+1)} h_j^{(L)}. \quad (23)$$

The FF neural network is estimated by applying the following formula:

$$x_{(k)}^{(l)} = f\left(x^{(l-1)'} \theta_k^{(l-1)}\right) \quad (24)$$

where, K^l refers to the number of concealed units in every layer $l = 1, \dots, L$, The output of the unit k in the layer l is symbolized by $x_{(k)}^{(l)}$, where $x^l = \left(1, x_{(1)}^{(l)}, \dots, x_{(k)}^{(l)}\right)'$ as the vector of outputs for the layer l , along with a constant term. The vector $x^{(0)}$ shows the input layer of the raw predictors. Every unit in the hidden layer follows the condition of $0 < l < L$ and is also frequently expressed in the above equation, and $x^{(l-1)'}$ shows the modified output vector from the preceding layer $l - 1$, and $\theta_k^{(l-1)}$ refers to the weight coefficients connecting the output of unit k to the layer L . The function f indicates the irregular activation of the function that modifies the weighted sum of the inputs. We apply the Rectified Linear Unit (ReLU) for the neural networks used in our paper:

$$f(u) = \begin{cases} 0 & \text{when } u < 0 \\ u & \text{when } u \geq 0 \end{cases} \quad (25)$$

The neural network's final result is computed using the equation below, which is similar to the linear output of the last secret layer.

$$g(z; \theta) = x^{(l-1)'} \theta_k^{(l-1)} \quad (26)$$

Following [21], we determine the turnover of the long-short portfolio at time t as follows:

$$TO_t = \frac{1}{2} \sum_{i \in L} \left| w_{i,t} - \frac{w_{i,t-1}(1 + re_{i,t})}{\sum_{k \in L} w_{k,t-1}(1 + re_{k,t})} \right| + \frac{1}{2} \sum_{j \in S} \left| w_{j,t} - \frac{w_{j,t-1}(1 + re_{j,t})}{\sum_{n \in S} w_{n,t-1}(1 + re_{n,t})} \right| \quad (27)$$

Where $i \in L$ ($j \in S$) displays that the coin i (j) takes the long (short) portfolio position, $w_{i,t}$ ($w_{j,t}$) shows the weight of coin i (j) at time t , $re_{i,t}$ ($re_{j,t}$)

Out-of-sample model estimation and assessment procedure

We modify the out-of-sample data set for each model's diurnal forecasts by partitioning the data into three mutually exclusive sets: the training, reference, and observational sets. First, we assess the control variable of the training sample TrS_1 across one year (February 14th, 2020, to February 14th, 2022), along with comprehensive hyperparameter maximisation of the 1-year reference sample TrS_2 (February 15th, 2022 to February 14th, 2023). Lastly, we perform an out-of-sample assessment of each model by applying the first 1-month t-test sample results to the extended data for the last lap (February 15th, 2024, to June 30th, 2025). In our analysis, the control variable is held fixed for one month, and the process is replicated by advancing the reference and observational sets by one month, while simultaneously extending the training sample dataset by one month in each recurrent loop. We carry out the process at the end of the sample data so that the complete dataset TrS_3 covers the period from February 14th, 2022, to June 30th, 2025. Our out-of-sample dataset analysis focuses on diurnal forecasts, as the short-rate sample period precludes weekly machine-learning estimation. The key performance indicator metric is the modified [42] of the sample R-squared (R_{oos}^2) numerator exercised by [21]:

$$R_{oos}^2 = 1 - \frac{\sum_{(i,t) \in TrS_3} (rre_{i,t+1} - rre_{i,t+1})^2}{\sum_{(i,t) \in TrS_3} (rre_{i,t+1})^2} \quad (28)$$

where, $rre_{i,t+1}$ and $rre_{i,t+1}$ depicts the realised returns at $t + 1$ and also for the succeeding day ahead $t + 1$ forecast for the crypto coin i , respectively, and TrS_3 states the metric function is evaluated only on the test sample.

The anomalies in cryptocurrency returns are evident in the very high kurtosis coefficient. The R-squared (R_{oos}^2) fact is based on the MSE, which assigns high weight to large errors; the anomalies can be significant. The out-of-sample MSE for the existing mean benchmark in relation to the competing forecast via the out-of-sample R^2 statistic [12], [7]

$$R_{i,oos}^2 = 1 - \frac{\sum_{s=1}^{TS-tini} (re_{i,tsini+s} - \widehat{re}_{i,tsini+s}^{CompeFore})^2}{\sum_{s=1}^{TS-tini} (re_{i,tsini+s} - \widehat{re}_{i,tsini+s}^{Benchmark})^2}, \quad (29)$$

Where, $\widehat{re}_{i,t}^{CompeFore}$ depicts a competing forecast, and t_{inis} (T) refers to the end of the initial sample period (i.e., the total sample). The proportional decrease in the out-of-sample MSE of the competing forecasts relative to the prevailing average benchmark. The [11] statistic, also known as the DMW statistic, is used to test whether a competing forecast yields a statistically significant decrease in MSE relative to the prevailing benchmark. The DMW statistics are calculated as the t-statistic, similar to the intercept a_i of the time-series regression as calculated below:

$$\underbrace{(re_{i,t} - \widehat{re}_{i,t}^{Benchmark})^2 - (re_{i,t} - \widehat{re}_{i,t}^{CompeFore})^2}_{D-Tlossdiff_{i,t}} = a_i + \varepsilon_{i,t} \text{ for } t = t_{ins} + 1, \dots, T \quad (30)$$

where, $D - Tlossdiff_{i,t}$ stands for the day- t loss differential, which covers the difference between the squared errors of the benchmark and the competing forecasts. The null hypothesis $H_0: a_i \leq 0$

Vector Autoregression (VAR) technique

Financial asset prices and order flows are closely linked and evolve together over time [16]. The positive correlation between them likely results from two factors: first, positive price insights encourage more buying activity, as order flow contains predictive information. Additionally, increased order flow exerts upward pressure on prices, creating a feedback loop that pushes prices higher. Alternatively, lower prices can attract more buy orders, which in turn draw in sell orders, thereby reducing overall trading velocity. The momentum of this cost push is quantitatively assessed as follows:

$$re_t = a_0 + \sum_{i=1}^5 a_i re_{t-1} + \sum_{i=0}^5 b_i OrFlow_{t-i}^{Global} + \sum_{i=1}^N m_i Ef_{i,t-1} + \varepsilon_{1,t} \quad (31)$$

$$OrFlow_t^{Global} = c_0 + \sum_{i=1}^5 c_i re_{t-1} + \sum_{i=0}^5 d_i OrFlow_{t-i}^{Global} + \sum_{i=1}^N n_i Ef_{i,t-1} + \varepsilon_{2,t} \quad (32)$$

here, $Ef_{i,t-1}$ is the $i \leq N$ refers to the economic variables applied as the control variables. For the selective unique constant and the cumulative factors at 5 lags, we compute the t-statistics using time- and coin-grouped standard errors. The sample period is February 14th, 2020, to June 30th, 2025.

ANALYSIS

Table 1-Analysis of the Predictive R^2 s for the average in-sample (AIS) and out-of-sample)

Model	(1)				(2)			
	$P_{Pct} = 50$		$P_{Pct} = 50$		$P_{Pct} = 50$		$P_{Pct} = 50$	
	AIS	OOS	AIS	OOS	AIS	OOS	AIS	OOS
OLS	7.60	1.24	8.29	-1.45	3.54	-4.82	4.49	-7.85
OLS+HL	7.58	1.35	8.26	-1.25	3.53	-4.70	4.46	-7.64
PCR	2.79	0.99	1.80	0.44	0.75	0.05	0.51	-0.05
PLS	6.34	3.58	6.29	2.83	1.12	-0.09	0.99	-0.27
LASSO	6.14	4.36	6.18	4.35	1.46	0.68	1.46	0.71
Lasso+HL	6.10	4.36	6.14	4.35	1.42	0.69	1.41	0.71
RIDGE	6.56	3.99	6.76	3.49	1.76	0.44	1.86	0.33
RIDGE+HL	6.52	3.99	6.71	3.44	1.73	0.45	1.83	0.35
ENET	6.14	4.36	6.18	4.35	1.45	0.68	1.45	0.71
ENET+HL	6.10	4.36	6.14	4.35	1.42	0.69	1.41	0.71
GLM	5.99	4.21	5.95	4.18	3.48	1.32	3.41	1.27
GLM+HL	5.95	4.22	5.98	4.19	3.42	1.34	3.34	1.30
RF	8.44	3.45	8.33	3.40	8.15	3.17	8.23	3.12
GBRT	7.18	3.45	7.12	3.44	6.61	2.86	6.52	2.94
GBRT+HL	7.26	3.55	7.21	3.47	6.57	3.22	6.47	3.33
NN1	6.63	4.47	6.82	4.38	5.71	2.88	5.90	2.69
NN2	6.66	4.52	6.82	4.36	6.32	3.23	6.43	2.99
NN3	6.57	4.44	6.76	4.37	6.13	2.99	6.19	2.78
NN4	6.57	4.41	6.76	4.34	5.99	2.91	6.14	2.61
NN5	6.51	4.37	6.66	4.24	5.91	2.82	5.80	2.30
ORACLE	6.33	5.62	6.33	5.62	5.96	5.50	5.96	5.50

Source: Author's Own Creation

Note- The above table displays the average in-sample (AIS) and out-of-sample (OOS) R^2 models (1) and (2) by applying the Ridge, Lasso, Elastic Net (ENet), summarised linear model with lasso group (GLM), random forest (RF), gradient boosted regression trees (GBRT), and five configurations of the neural networks (NN1, ..., NN5), respectively. “+HL” refers to the Huber loss instead of the l_2 loss, Oracle refers to the current covariates used in the pooled OLS regression. We set the no. of the sample at 200, where $N=2$, $T=180$, and $P_{Pct} = 2$ in comparison to the $P_{Pct} = 10$ with $P_{Pct} = 50$ And also the number of Monte Carlo repetitions is 100.

The average in-sample (AIS) and out-of-sample (OOS) R^2 models relate to estimators based on the AIS average. For model (1), Lasso, ENet, and NNs show superior and nearly equal OOS R^2 results, though high-performing models are rare and vary with the study's inlet covariates. Advanced tree methods like RF and GBRT tend to overfit, so they perform slightly worse. In contrast, in model (2), these methods clearly outperform Lasso and ENet, as the latter fail to capture the models' nonlinearities. The GLM performs similarly well, although it is driven by NNs, RF, and GBRT. OLS results are the worst across all conditions, while PLS outperforms PCR in linear model (1) and exceeds it in nonlinear models. As the P_{Pct} increases, AIS R^2 rises, but OSS R^2 declines, indicating a deterioration in performance as selection bias increases. Applying the Huber Loss improves OOS performance for nearly all methods, with RF, GBRT, and HL considered optimal for nonlinear models. The NNs show a notable trade-off between flexibility and complexity; the ENet's Accelerated Proximal Gradient (APG) algorithms are unsuitable for all NNs due to non-linear loss functions. The table notes that lower NNs outperform higher ones.

Table 2 Analysis of the Predictive R^2 s for the average in-sample (AIS) and out-of-sample) for different prediction time periods in Simulations

Model	(1)						(2)					
Range	Qtr.		HYrly		Yearly		Qtr.		H.Yr.		Yearly	
R^2 (%)	AIS	OOS	AIS	OOS	AIS	OOS	AIS	OOS	AIS	OOS	AIS	OOS
OLS	18.94	-0.95	27.77	0.29	35.50	-0.25	10.13	-16.57	15.22	-23.58	20.29	-30.83
OLS+HL	18.92	-0.86	27.76	0.41	35.48	-0.19	10.10	16.37	14.95	-23.42	20.25	-30.76
PCR	3.96	0.95	5.60	1.42	7.68	1.49	0.99	-0.14	1.40	-0.14	1.81	-0.34
PLS	15.22	6.66	21.62	8.50	26.56	8.24	1.95	-0.52	1.83	-0.43	2.88	-0.90
LASSO	14.20	10.43	20.52	14.78	25.16	16.76	3.20	1.27	4.13	1.22	4.97	0.50
Lasso+HL	14.21	10.42	20.40	14.77	24.95	16.84	3.13	1.29	3.95	1.25	4.76	0.58
RIDGE	15.86	7.91	23.36	10.77	29.37	11.75	4.17	0.53	5.76	0.54	6.82	0.00
RIDGE+HL	15.78	7.94	23.25	10.79	29.23	11.75	4.10	0.55	5.66	0.56	6.66	0.14
ENET	14.18	10.43	20.59	14.79	25.16	16.79	3.20	1.25	4.17	1.24	4.90	0.51
ENET+HL	13.95	10.42	20.47	14.78	24.95	16.76	3.12	1.27	3.95	1.28	4.70	0.58
GLM	13.99	9.50	21.12	13.63	27.25	15.46	7.71	2.56	10.89	2.99	13.17	1.73
GLM+HL	13.89	9.52	20.98	13.66	26.95	15.50	7.58	2.61	10.69	2.99	12.88	1.81
RF	17.66	8.21	25.34	11.96	31.14	14.42	15.62	5.95	20.63	7.21	22.58	6.15
GBRT	15.99	8.95	22.78	13.37	28.78	15.16	12.49	5.97	15.95	6.99	18.18	5.95
GBRT+HL	15.80	8.88	22.94	13.55	29.17	15.39	12.22	5.97	16.10	7.23	18.30	6.27
NN1	15.78	9.99	23.14	14.17	29.72	15.68	13.32	5.46	17.99	6.39	20.78	5.42
NN2	15.66	9.96	22.82	14.10	28.99	16.11	13.39	5.86	17.99	6.87	20.20	5.53
NN3	15.55	9.89	22.77	13.95	28.69	16.20	13.21	5.67	17.60	6.73	20.41	5.37
NN4	15.59	9.95	22.42	14.16	28.69	15.99	13.30	5.66	17.90	6.62	19.77	5.30
NN5	15.29	9.92	22.24	13.95	28.32	15.95	13.10	5.34	17.25	6.29	18.96	5.18
ORACLE	14.47	12.82	20.83	18.25	25.52	21.66	10.99	10.38	13.71	12.85	13.14	11.62

Source: Author's Own Creation

Note- The above table displays the average in-sample (AIS) and out-of-sample (OOS) R^2 models (1) and (2) by applying the Ridge, Lasso, Elastic Net (ENet), summarised linear model with lasso group (GLM), random forest (RF), gradient boosted regression trees (GBRT), and five configurations of the neural networks (NN1, ..., NN5), respectively. “+HL” refers to the Huber loss instead of the l_2 loss, Oracle refers to the current covariates used in the pooled OLS regression. We set the no. of the sample at 200, where $N=2$, $T=180$, and $P_{pct} = 2$ in comparison to the $P_{pct} = 10$ with $P_{pct} = 50$ And also the number of Monte Carlo repetitions is 100.

The usual ever-increasing hump-shaped pattern of the R^2 s Against varied investment horizons, it is specifically driven by the covariate's persistence, with comparable performance across the models.

Table 3 Comparison of the Mean Variable Selection Frequencies in Simulation

Model (1)								
Parameter	Technique	$cv_{i,1,t}$	$cv_{i,2,t}$	$cv_{i,3,t}$	$cv_{i,1,t} \times x_t$	$cv_{i,2,t} \times x_t$	$cv_{i,3,t} \times x_t$	Dissonance
$P_{pct} = 50$	Lasso	0.96	0.95	0.66	0.54	0.52	0.86	0.10
	Lasso +HL	0.96	0.95	0.64	0.54	0.51	0.87	0.09
	ENet	0.96	0.95	0.66	0.55	0.52	0.87	0.10
	ENet+HL	0.96	0.95	0.65	0.54	0.51	0.87	0.10
	GLM	0.96	0.95	0.73	0.62	0.64	0.91	0.14
	GLM+ HL	0.96	0.95	0.71	0.62	0.63	0.91	0.13
$P_{pct} = 100$	Lasso	0.96	0.95	0.66	0.53	0.50	0.86	0.07
	Lasso +HL	0.96	0.95	0.64	0.54	0.50	0.87	0.07
	ENet	0.96	0.95	0.66	0.54	0.50	0.87	0.07
	ENet+HL	0.96	0.95	0.65	0.54	0.50	0.87	0.07
	GLM	0.96	0.95	0.73	0.59	0.62	0.91	0.10

	GLM+ HL	0.96	0.95	0.70	0.56	0.61	0.91	0.10
Model (2)								
Parameter	Technique	$cv_{i,1,t}$	$cv_{i,2,t}$	$cv_{i,3,t}$	$cv_{i,1,t} \times x_t$	$cv_{i,2,t} \times x_t$	$cv_{i,3,t} \times x_t$	Dissonance
$P_{Pct} = 50$	Lasso	0.27	0.27	0.40	0.28	0.32	0.76	0.05
	Lasso +HL	0.26	0.26	0.39	0.29	0.32	0.75	0.05
	ENet	0.27	0.26	0.40	0.28	0.32	0.7	0.05
	ENet+HL	0.26	0.25	0.40	0.29	0.32	0.76	0.05
	GLM	0.81	0.55	0.69	0.69	0.65	0.83	0.22
	GLM+ HL	0.80	0.55	0.71	0.69	0.63	0.83	0.21
$P_{Pct} = 100$	Lasso	0.26	0.26	0.38	0.26	0.32	0.76	0.03
	Lasso +HL	0.25	0.25	0.37	0.27	0.32	0.76	0.03
	ENet	0.26	0.26	0.38	0.26	0.32	0.77	0.03
	ENet+HL	0.25	0.25	0.38	0.27	0.32	0.76	0.03
	GLM	0.81	0.53	0.68	0.66	0.58	0.82	0.15
	GLM+ HL	0.80	0.49	0.68	0.67	0.57	0.82	0.14

Source: Author's Own Creation

Note- The above table reports the mean variable selection for the frequencies of the 6 particulate variables for the 2 models (1) and (2) covering the monthly horizon by applying the Lasso, Elastic Net (ENet) and the generalised linear model with the group lasso (GLM), respectively. “+HL” refers to the Huber loss instead of the l_2 loss. “Dissonance” states the mean selection frequency of the remaining P-6 covariates. We set the no. of the sample at 200, where $N=2$, $T=180$, and $P_{Pct} = 2$ in comparison to the $P_{Pct} = 10$ with $P_{Pct} = 50$ and also the number of Monte Carlo repetitions is 100.

We use the Lasso, Elastic Net and Lasso Group along with their most powerful versions, as they all enforce the $loss_1$ penalty and thereby select the variable selection. The true covariates ($cv_{i,1,t}$, $cv_{i,2,t}$, $cv_{i,3,t} \times x_t$) are selected in over 85% of the sample size, unlike the superfluous covariates ($cv_{i,3,t}$, $cv_{i,1,t} \times c$, $cv_{i,2,t} \times x_t$) are selected in around 60% of the samples. The last covariate is occasionally chosen. The pressure on variable selection and prediction, or on finite-sample issues, is that the true covariates are included in the selected models with very high probability. There are zero covariates for the second model; the covariates included are far more useful and therefore are selected substantially more often than the rest of the covariates $P - 6$ covariates.

Table 4 Comparison of the Mean Variable Selection Frequencies in Simulation

Model (1)								
Parameter	Technique	$cv_{i,1,t}$	$cv_{i,2,t}$	$cv_{i,3,t}$	$cv_{i,1,t} \times x_t$	$cv_{i,2,t} \times x_t$	$cv_{i,3,t} \times x_t$	Dissonance
$P_{Pct} = 50$	RF	22.55	23.30	5.99	6.55	6.45	19.55	0.20
	GBRT	24.68	27.88	5.74	6.46	6.13	25.77	0.06
	GNRT+HL	24.03	24.63	5.53	6.50	6.88	25.99	0.06
	NN1	26.60	29.99	5.61	2.88	3.76	25.28	0.08
	NN2	26.45	28.96	5.24	3.23	3.96	25.84	0.08
	NN3	26.15	28.67	5.23	3.19	3.89	25.67	0.09
	NN4	26.29	28.92	5.28	3.47	3.86	25.69	0.09
	NN5	25.96	28.60	5.32	3.36	3.76	25.46	0.10
	NN6	25.96	28.60	5.32	3.36	3.76	25.46	0.10
$P_{Pct} = 100$	RF	21.60	23.18	5.94	5.63	5.89	19.28	0.11
	GBRT	24.87	28.92	5.52	6.23	5.89	25.53	0.04
	GNRT+HL	23.85	27.22	5.40	6.14	6.36	26.10	0.04
	NN1	25.88	28.56	5.23	2.88	3.63	24.67	0.06
	NN2	25.60	27.99	4.95	2.96	3.55	24.68	0.07
	NN3	25.82	28.33	4.83	2.99	3.56	24.63	0.07

	NN4	25.22	27.73	4.88	2.93	3.48	24.44	0.07
	NN5	24.89	27.93	4.86	3.17	3.53	24.29	0.07
Model (2)								
Parameter	Technique	$cv_{i,1,t}$	$cv_{i,2,t}$	$cv_{i,3,t}$	$cv_{i,1,t} \times x_t$	$cv_{i,2,t} \times x_t$	$cv_{i,3,t} \times x_t$	Dissonance
$P_{Pct} = 50$	RF	27.80	6.49	5.06	8.12	5.10	32.23	0.19
	GBRT	31.25	7.38	5.92	8.82	6.53	36.51	0.14
	GNRT+HL	32.15	7.47	5.93	8.88	6.72	35.67	0.14
	NN1	55.66	14.50	4.56	3.56	2.98	12.21	0.17
	NN2	51.94	13.77	4.25	2.97	2.82	18.33	0.17
	NN3	52.10	13.74	4.46	2.95	2.99	16.73	0.18
	NN4	51.09	13.71	4.55	3.36	2.89	16.29	0.19
	NN5	49.76	13.78	4.58	3.38	2.96	15.99	0.21
$P_{Pct} = 100$	RF	26.43	5.84	4.60	7.78	4.79	31.96	0.11
	GBRT	31.50	7.40	5.48	8.67	6.27	36.80	0.05
	GNRT+HL	32.23	7.58	5.81	8.77	6.40	35.89	0.05
	NN1	53.11	13.63	4.89	3.55	2.88	12.12	0.06
	NN2	50.26	12.96	4.36	2.88	2.55	17.41	0.06
	NN3	50.46	13.10	4.40	2.91	2.64	15.46	0.07
	NN4	48.38	13.15	4.60	3.23	2.73	15.15	0.08
	NN5	43.44	12.51	4.81	3.63	2.69	16.19	0.10

Source: Author's Own Creation

Note- The above table reports the mean variable selection for the frequencies of the 6 particulate variables for the 2 models (1) and (2) covering the monthly horizon by applying Random Forest (RF), gradient boosted regression trees (GBRT), and five architectures of neural networks (NN1,...NN), respectively. “+HL” refers to the Huber loss instead of the l_2 loss. “Dissonance” sates the mean selection frequency of the remaining P-6 covariates. We set the no. of the sample at 200, where N=2, T=180, and $P_{Pct} = 2$ in comparison to the $P_{Pct} = 10$ with $P_{Pct} = 50$ and also the number of Monte Carlo repetitions is 100. The above table reports the mean VIPs of the specific covariates and the mean of the remaining $P - 6$

Table 5 Variance segregation of the global order flows

	Diurnal%	Per Week%
$OrFlow^{USD}$	38.07	38.13
$OrFlow^{KRW}$	23.43	30.49
$OrFlow^{EUR}$	10.47	10.08
$OrFlow^{GBP}$	5.99	4.89
$OrFlow^{NZD}$	3.88	3.23
$OrFlow^{CHF}$	3.75	2.49
$OrFlow^{AUD}$	3.67	2.66
$OrFlow^{CAD}$	3.54	2.88
$OrFlow^{NOK}$	2.84	1.88
$OrFlow^{SEK}$	2.76	1.61
$OrFlow^{JPY}$	1.60	1.66
$OrFlow^{World}$	100	100

Source: Author's Own Creation

Table 2 presents the variance decomposition of world order flows across 11 countries where cryptocurrencies are primarily traded. The daily data show the highest variance contribution at 38%; South Korea ranks second at 23%, making it one of the elite players in the crypto market. The Euro ranked third at 11%, followed by the British Pound at about 6%. Broadly, the world order flow is dominated by a few countries: the top 2 countries account for almost 61% of the total variance, and the top 3 account for approximately 72%. The weekly variance contributions are similar to the daily ones, regardless of the order flow of the top three countries, which together account for 79% of global order flow. The values represent the percentage contribution to the variance of each

country's world order flows at daily and weekly frequencies. The out-of-sample testing period runs from February 14th, 2020, to June 30th, 2025.

Table 6 Concurrent panel regressions with the Global Order Flows

	Diurnal		Per Week	
	(1)	(2)	(3)	(4)
Constant term	0.003	0.002	0.013	0.060
	(1.20)	(1.17)	(1.51)	(1.18)
$OrFlow_{i,t}^W$	0.033***	0.033***	0.055***	0.050***
	(15.78)	(15.70)	(8.20)	(8.39)
$MktCapi_{i,t}$		-0.002		0.020***
		(-0.60)		(3.18)
$TotalVol_{i,t}$		0.008***		0.008***
		(6.96)		(2.58)
$NlogVol_{i,t}$		0.269**		0.266***
		(3.64)		(3.42)
$CBOEVIX_t$		-0.050		-0.095
		(-1.62)		(-1.18)
$FRED_T$		0.000		0.001
		(0.89)		(0.88)
$S\&P\ 500\ VIX_t$		-0.677		-4.066
		(-2.06)		(-2.30)
$MSCI_t^{World}$		0.784		5.892***
		(1.44)		(2.86)
$ShRate_t$		-0.003		-0.018
		(-0.86)		(-1.46)
$Yield\ Spread_t$		-0.008		-0.066**
		(-1.51)		(-1.99)
$Z(default)Spread_t$		0.011		0.014
		(1.37)		(0.30)
R-Squared (%)	7.1	11.36	6.7	21.4
No. of Obs.	101139	101139	18723	18723

Source: Author's Own Creation

The table above presents the results of the concurrent panel regression, which includes global order flow and various control variables. The study uses the log version of order flow, with market capitalisation and total volatility also expressed in logarithmic form. “Volatility” indicates the difference between a cryptocurrency’s high and low prices. Log returns are used for the CBOEVIX, S&P 500 VIX, and MSCI World indices. FRED presents the difference between the 3-month Treasury bill rate and LIBOR, with the short rate denoting the former. The yield spread measures the difference between the 10-year and 3-month Treasury bill rates. The Z-default spread is the difference between AAA and BAA bonds. The total panel regression is based on a balanced panel that includes fixed effects for each cryptocurrency. Columns (1) and (2) show daily returns, while columns (3) and (4) show weekly returns. Statistics are in parentheses and are calculated using standard errors, grouped by time and coins. The out-of-sample testing period runs from February 14th, 2020, to June 30th, 2025.

The table shows that global order flow has a marked impact on cryptocurrency returns, a pattern observed across all cases. This indicates that global order flows effectively account for variations in returns. For instance, a 1% increase in the standard deviation of control variables, such as global order flows, is associated with a 3.3% increase in daily returns and a 5.0% increase in weekly returns. These results are supported by high t-statistics of 15.70 for daily data and 8.39 for weekly data, confirming their robustness. Our findings are consistent with [21], who also documented a positive relationship. The analysis establishes a contemporaneous link between global order flow and Bitcoin returns in daily regressions, with this positive association extending to other cryptocurrencies and to weekly models. The weekly regressions show higher coefficients than the daily ones, with R-squared values of 11.36% for daily and

21.4% for weekly data. The stronger weekly results likely stem from order-flow data that filter out high-frequency noise and uninformed trades. Summing order flow over a week reduces this noise, which may explain its greater explanatory power for weekly price changes. Besides order flow, other control variables are also significant: for daily data, total volume and volatility; for weekly data, total market cap, volume, volatility, S&P 500 VIX returns, MSCI Global index returns, and yield spreads. Overall, these results demonstrate that order flows have substantial explanatory power for cross-sectional cryptocurrency returns, even after controlling for crypto-specific factors and basic economic indicators.

Table 7 Predictive longitudinal data analysis with the global order flows

	Diurnal			Per Week		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant term	0.003	0.003	0.014	0.014	0.014	0.056
	(1.03)	(1.09)	(1.09)	(1.52)	(1.54)	(0.78)
$OrFlow_{i,t-1}^W$	0.002	0.002***	0.002***	0.009**	0.009***	0.009***
	(0.72)	(2.68)	(3.12)	(2.40)	(3.33)	(3.50)
$re_{i,t-1}$		-0.064***	-0.067		-0.029	-0.038
		(-3.35)	(-3.83)		(-0.52)	(-1.02)
$MktCap_{i,t-1}$			-0.008***			-0.031***
			(-3.85)			(-6.00)
$TotalVol_{i,t-1}$			0.002			0.005**
			(1.49)			(1.09)
$NlogVol_{i,t-1}$			0.058***			0.077*
			(2.74)			(3.08)
$CBOEVIX_{t-1}$			-0.014			-0.106
			(-0.40)			(-1.29)
$FRED_{t-1}$			-0.000			-0.001
			(0.02)			(-0.40)
$S\&P\ 500\ VIX_{t-1}$			-0.150			1.575
			(-0.30)			(1.60)
$MSCI_{t-1}^{World}$			0.498			-1.988
			(0.88)			(-1.88)
$ShRate_{t-1}$			-0.007*			-0.025
			(-1.82)			(-1.62)
$Yield\ Spread_{t-1}$			-0.005			-0.007
			(-0.51)			(-0.18)
$Z(default)Spread_t$			-0.006			-0.007
			(-0.48)			(-0.12)
R-Squared (%)	0.1	0.3	1.3	0.2	0.2	3.6
No. of Obs.	101047	101047	101047	18631	18631	18631

Source: Author's Own Creation

The table above presents the predictive longitudinal data analysis of cryptocurrency returns, along with lagged global order flow and lagged control factors. The table shows the same information as concurrent panel regressions, but all the factor variables are lagged and $re_{i,t-1}$ refers to the lagged return of each cryptocurrency under study. Statistics are in parentheses and are calculated using standard errors, grouped by time and coins. The out-of-sample testing period runs from February 14th, 2020, to June 30th, 2025.

The table's results show that global order flows are a highly significant predictor of cryptocurrency returns both 1 day and 1 week ahead. The Diurnal predictive regression shows an unusual pattern: lagged global order flows are not significant on their own (t-stat = 0.72), but when controlling for other factors, their logged returns become highly significant (t-stat = 2.68). Conversely, the effect of the logged returns is negative and highly significant (t-stat = -3.35). This suggests that lagged order flows have two opposing influences: a short-term

reversal associated with lagged returns and a long-term factor that, while uncorrelated with lagged returns, is positively linked to future crypto returns. Other variables, including total market capitalisation, volatility, and the short rate, are important for the daily regressions. For weekly regressions, market capitalisation, MSCI global uncertainty, and volatility are key factors. The R-squared values are 1.3% for the daily regression and 3.6% for the weekly regression.

Table 8 Concurrent Panel regression with all multinational order flows

	Diurnal		Per Week	
	(1)	(2)	(3)	(4)
Constant term	0.003	0.002	0.063	0.060
	(0.20)	(0.009)	(1.29)	(1.26)
$OrFlow_{i,t}^W$		0.021***		0.031***
		(13.03)		(6.96)
$OrFlow_{i,t}^{USD}$	0.014***	0.007***	0.028***	0.016***
	(8.88)	(4.50)	(6.66)	(4.28)
$OrFlow_{i,t}^{EUR}$	0.013***	0.008***	0.020***	0.009***
	(7.76)	(6.33)	(5.88)	(3.82)
$OrFlow_{i,t}^{KRW}$	0.005***	0.001	0.0011**	0.004*
	(5.25)	(0.76)	(3.99)	(1.84)
$OrFlow_{i,t}^{GBP}$	0.003***	-0.001**	0.006***	-0.002
	(4.19)	(-2.40)	(2.79)	(-0.50)
$OrFlow_{i,t}^{JPY}$	0.001*	-0.001	0.002	-0.002
	(1.85)	(-0.63)	(0.85)	(-0.40)
$OrFlow_{i,t}^{CAD}$	0.001***	-0.001***	0.006***	0.003
	(2.71)	(-2.73)	(4.26)	(1.29)
$OrFlow_{i,t}^{AUD}$	0.001	-0.002***	0.002*	-0.002
	(1.55)	(-3.19)	(1.88)	(-0.56)
$OrFlow_{i,t}^{CHF}$	0.001	-0.002***	0.002	-0.002*
	(0.82)	(-6.73)	(0.36)	(-1.84)
$OrFlow_{i,t}^{NZD}$	0.002	-0.002***	-0.002	-0.003*
	(1.69)	(-5.73)	(-0.46)	(-3.40)
$OrFlow_{i,t}^{NOK}$	0.001	-0.002***	0.002	-0.002
	(0.56)	(-5.11)	(1.07)	(-0.36)
$OrFlow_{i,t}^{SEK}$	0.001	-0.002***	0.002	-0.002
	(0.84)	(-5.26)	(0.83)	(-0.92)
R-Squared (%)	9.8	12.5	20.8	22.2
No. of Obs.	101139	101139	18723	18723

Source: Author's Own Creation

The table shows a concurrent panel regression of cryptocurrency returns on global order flows, multinational order flows, and control factors. It also presents the combined analysis for all multinationals alongside global order flows. Statistics are shown in parentheses, calculated with standard errors grouped by time and coins. The out-of-sample testing covers the period from February 14th, 2020, to June 30th, 2025.

Table 9 Exponential Smoothing of Panel Regression—Nonlinear factors test sample period

Single Predictive Panel Regression				
	Linear	Square	Cubic	Interrelation with $OrFlow_{i,t}^W$
Constant	0.0293			
	(1.27)			
$OrFlow_{i,t-1}^W$	0.0044**	-0.0009*	-0.0003*	
	(3.50)	(-1.96)	(-1.85)	

$OrFlow_{i,t-1}^{USD}$	-0.0001	0.0002	0.0001	0.0005
	(-0.10)	(0.78)	(1.09)	(1.00)
$OrFlow_{i,t-1}^{EUR}$	-0.0007	-0.0002	0.0001	-0.0001
	(-1.18)	(-0.64)	(0.46)	(-0.27)
$OrFlow_{i,t-1}^{KRW}$	0.0016**	0.0003	0.0001	0.0015***
	(2.20)	(1.06)	(1.40)	(2.79)
$OrFlow_{i,t-1}^{GBP}$	-0.0007	-0.0001	0.0001	0.0000
	(-1.17)	(-0.44)	(0.77)	(0.10)
$OrFlow_{i,t-1}^{JPY}$	0.0002	-0.001	-0.0001*	-0.0000
	(0.49)	(-0.34)	(-1.93)	(0.04)
$OrFlow_{i,t-1}^{CAD}$	-0.0005	0.0003	-0.0002	-0.0002
	(-0.83)	(0.77)	(-1.25)	(-0.55)
$OrFlow_{i,t-1}^{AUD}$	-0.0008	0.0006**	0.0000	-0.0006
	(-1.59)	(1.99)	(0.42)	(-1.07)
$OrFlow_{i,t-1}^{CHF}$	-0.0004	-0.0009***	0.0001	0.0005
	(-0.60)	(-2.93)	(1.55)	(1.27)
$OrFlow_{i,t-1}^{NZD}$	-0.0003	0.0005	0.0000	-0.0002
	(-0.56)	(1.13)	(0.60)	(-0.30)
$OrFlow_{i,t-1}^{NOK}$	-0.0002	0.0008**	-0.0000	-0.0004
	(-0.33)	(2.44)	(-0.20)	(-0.78)
$OrFlow_{i,t-1}^{SEK}$	-0.0001	0.0004	-0.0002	-0.0009
	(-0.21)	(0.98)	(-1.08)	(-1.58)
$re_{i,t-1}$	-0.0852***	0.0450	-0.0097	0.0010
	(-3.03)	(0.88)	(-0.64)	(1.43)
F-test (non-linear)		0.068	0.000	0.096
F-test (All non-linear)		0.000		
R^2 (%) (Discrete regression)	1.99	2.14	2.21	2.29
No. of observations	101139	101139	18723	18723

Source: Author's Own Creation

The table shows the results of a specific predictive regression of cryptocurrency returns on the linear, squared, cubic, and cross-product terms of the lagged world order flow and international order flows. The exponential smoothing of the panel regressions also accounts for the control variables estimated in the previous tables, but these results are not reported to save space. The t-statistics in parentheses are calculated using the standard errors grouped by time and coins. The adjusted $\bar{R}^2$ values are reported at the bottom, along with the four individual regressions: the first conditioned only on the linear terms, the second also including squared terms, the third adding cubic terms, and the last incorporating all terms. The out-of-sample testing period runs from February 14th, 2020, to June 30th, 2025.

Table 10 Vector Autoregression Result: BTC and XRP

	BTC				XRP			
	Returns		Order Flow		Returns		Order Flow	
	Parameter	(t-stat)	Parameter	(t-stat)	Parameter	(t-stat)	Parameter	(t-stat)
Constant	0.013	0.99	0.264	0.99	-0.010	(-0.72)	0.534*	(2.09)
$OrFlow_{i,t}^W$	0.025***	(13.05)			0.047***	(12.97)		
$OrFlow_{i,t-1}^W$	0.002	(1.16)	0.060	(1.52)	0.007**	(2.57)	0.017	(0.27)
$OrFlow_{i,t-2}^W$	-0.001	(-0.84)	0.093**	(3.04)	0.007*	(1.99)	-0.039	(-0.89)
$OrFlow_{i,t-3}^W$	-0.001	(-0.66)	0.017	(0.45)	0.005*	(1.99)	0.055	(1.10)
$OrFlow_{i,t-4}^W$	0.000	(0.06)	0.059	(1.49)	-0.005	(-1.58)	-0.009	(-0.19)

$OrFlow_{i,t-5}^W$	0.005*	(1.96)	0.033	(0.97)	0.001	(0.46)	0.026	(0.76)
$re_{i,t-1}$	-0.037	(-2.27)	-3.330***	(-3.83)	-0.028	(-0.83)	-1.188*	(-1.78)
$re_{i,t-2}$	0.043	(1.29)	0.395	(0.39)	-0.097**	(-3.44)	1.376**	(3.31)
$re_{i,t-3}$	0.025	(0.80)	1.749**	(2.05)	-0.042	(-1.00)	-0.040	(-0.09)
$re_{i,t-4}$	-0.013	(0.34)	0.206	(0.35)	0.073*	(1.84)	0.994*	(1.85)
$re_{i,t-5}$	-0.088*	(-1.94)	-1.068	(-1.27)	-0.009	(17)	-0.085	(-0.26)
$MCAP_{i,t-1}$	-0.005	(-0.63)	-0.067	(-0.58)	-0.019**	(-2.50)	-0.365***	(-3.70)
$Volume_{i,t-1}$	-0.006	(-0.88)	-0.008	(-0.08)	0.006	(0.60)	-0.068	(-0.98)
$Volatility_{i,t-1}$	0.086**	(2.22)	0.497	(0.56)	0.050	(0.98)	1.534***	(3.95)
VIX_{t-1}	0.037	(1.57)	-0.590	(-0.99)	-0.050	(-1.37)	-0.554	(-0.96)
$TED\ Spread_{t-1}$	0.000	(0.10)	-0.001	(0.75)	0.00	(-0.96)	-0.001	(-0.76)
$S\&P500_{t-1}$	0.412	(0.86)	-11.7000	(-2.48)	-0.795*	(-1.93)	3.925	(0.52)
$MSCI_{t-1}^{Global}$	-0.080	(-0.25)	10.956	(1.34)	0.970**	(2.24)	-2.506	(-0.25)
$Short\ rate_{t-1}$	-0.008*	(-2.55)	-0.082	(-0.93)	0.006	(0.86)	-0.092	(-1.00)
$Term\ Spread_{t-1}$	-0.008	(-2.10)	-0.019	(-0.16)	0.019*	(1.88)	0.030	(0.30)
$Default\ Spread_{t-1}$	0.000	(0.05)	-0.142	(-0.69)	-0.006	(-0.64)	-0.552***	(-3.82)
R^2 (%)	30.88		1.81		47.81		2.24	

Source: Author's Own Creation

The table above presents vector autoregressions (VARs) of cryptocurrency returns and order flow for two coins: Bitcoin (BTC) and XRP (Ripple). The VAR stipulations follow [25]. T-statistics shown in parentheses are computed using the standard errors that are clustered both by time and coin. The out-of-sample testing period runs from February 18, 2020, to June 30, 2025.

Table 11 Portfolios sorted based on the U.S. order flow

Part A: Daily Portfolios Sorted based on US order Flow								
	Pr_1	Pr_2	Pr_3	Pr_4	Pr_5	$Pr_5 - Pr_1$		
	Mean Returns (% Daily)					Mean	Alpha	SR
$OrFlow^{USD}$	0.51 (2.15)	0.48 (1.90)	0.43 (1.70)	0.46 (1.85)	0.43 (1.721)	-0.09 (-0.81)	-0.09 (-0.75)	-0.49
Ortho- $OrFlow^{USD}$	0.49 (2.00)	0.47 (1.90)	0.33 (1.34)	0.50 (2.05)	0.50 (2.03)	0.01 (0.10)	0.03 (0.27)	0.07
Part B: Daily Portfolios Sorted based on US order Flow								
	Pr_1	Pr_2	Pr_3	Pr_4	Pr_5	$Pr_5 - Pr_1$		
	Mean Returns (% Daily)					Mean	Alpha	SR
$OrFlow^{USD}$	1.18 (0.91)	2.56 (1.84)	2.00 (1.55)	2.50 (1.85)	3.02 (2.29)	1.85 (3.16)	1.78 (2.84)	1.95
Ortho- $OrFlow^{USD}$	1.05 (0.78)	2.26 (1.75)	2.51 (1.79)	2.61 (1.98)	2.85 (2.16)	1.80 (3.12)	1.61 (2.72)	1.98

Source: Author's Own Creation

The table above shows the performance of the cryptocurrency portfolios classified by the U.S. order flow $OrFlow^{USD}$. $Ortho - OrFlow^{USD}$ refers to the US order flow that is orthogonal to the same-period returns. Pr_1 refers to the portfolio with the lowest lagged order flow, and Pr_5 shows the portfolio encompassing the highest lagged order flow. The portfolios are equally weighted and recalculated daily (Part A) or weekly (Part B). The alpha is explained by a three-factor model comprising the cryptocurrency market, size, and momentum factors [30]. The returns and the alphas are converted into daily/weekly percentages. SR refers to the annual Sharpe Ratio. The Newey and West (1987) *t-statistics* are shown in parentheses. The out-of-sample testing period runs from February 18, 2020, to June 30, 2025.

Table 12 Daily portfolios classified on ML forecasts

	OF Model			EF Model			OF + EF model		
	$P_5 - P_1$			$P_5 - P_1$			$P_5 - P_1$		
	Mean	Alpha (α)	ASR	Mean	Alpha (α)	ASR	Mean	Alpha (α)	ASR
OLS (t-stat)	-0.18 (-1.04)	-0.17 (-0.98)	-0.68	-0.04 (-0.13)	0.04 (0.11)	-0.09	-0.11 (-0.56)	-0.08 (-0.41)	-0.36
RR (t-stat)	-0.15 (-0.87)	-0.15 (-0.83)	-0.57	-0.09 (-0.44)	-0.06 (-0.22)	-0.28	-0.13 (-0.72)	-0.10 (-0.58)	-0.46
LAS (t-stat)	-0.20 (-1.29)	-0.20 (-1.25)	-0.83	0.13 (0.98)	0.13 (0.95)	0.62	0.14 (1.10)	0.15 (1.07)	0.69
EN (t-stat)	-0.15 (-0.88)	-0.13 (-0.72)	-0.56	-0.03 (-0.06)	0.04 (0.14)	-0.04	0.00 (-0.10)	0.02 (0.16)	-0.01
PCR (t-stat)	-0.07 (-0.34)	-0.06 (-0.30)	-0.22	0.00 (0.02)	-0.06 (-0.25)	0.03	-0.04 (-0.17)	-0.09 (-0.45)	-0.11
RF (t-stat)	0.42 (2.59)	0.44 (2.82)	1.74	-0.23 (-1.46)	-0.21 (-1.31)	-0.93	-0.12 (-0.70)	-0.13 (-0.74)	-0.45
SGB (t-stat)	0.55 (3.27)	0.58 (3.50)	2.16	0.24 (1.92)	0.23 (1.91)	1.22	0.20 (1.61)	0.21 (1.63)	1.03
NN1 (t-stat)	0.13 (0.71)	0.17 (0.95)	0.45	-0.29 (-1.69)	-0.25 (-1.39)	-1.03	-0.15 (-0.84)	-0.12 (-0.60)	-0.53
NN2 (t-stat)	-0.07 (-0.31)	0.00 (0.02)	-0.20	-0.02 (-0.11)	0.02 (0.10)	-0.07	-0.10 (-0.59)	-0.06 (-0.28)	-0.39
NN3 (t-stat)	-0.05 (-0.24)	-0.04 (-0.12)	-0.16	-0.12 (-0.70)	-0.10 (-0.63)	-0.47	-0.19 (-1.05)	-0.15 (-0.82)	-0.67
NN4 (t-stat)	-0.08 (-0.44)	-0.07 (-0.30)	-0.27	-0.14 (-0.90)	-0.09 (-0.51)	-0.53	-0.09 (-0.53)	-0.04 (-0.21)	-0.35
L-Mean (t-stat)	0.00 (0.00)	0.00 (0.03)	0.00	-0.09 (-0.49)	-0.05 (-0.22)	-0.30	-0.09 (-0.47)	-0.06 (-0.28)	-0.29
NL-Mean (t-stat)	0.44 (2.80)	0.48 (2.96)	1.79	-0.38 (-2.21)	-0.34 (-1.83)	-1.36	-0.19 (-1.11)	-0.15 (-0.82)	-0.71

Source: Author's Own Creation

The table above presents the performance of a cryptocurrency portfolio, segmented by Machine Learning forecasts, using the top 20 cryptocurrencies by market cap as of February 15, 2020. ML techniques are applied to a full cross-sectional sample of 128 coins. The top portfolio ($Port_3$) includes the five coins with the highest ML forecasts, while the bottom portfolio ($Port_3$) contains the coins with the lowest forecasts. Both portfolios are equally weighted and rebalanced daily. The study analyses three ML models: one where order flow depends only on itself; another where EF depends on economic indicators; and a third where OF+EF jointly constrain both. All models depend on lagged returns. The alpha is computed using a three-factor model including the crypto market, size, and momentum (Liu, Tsyvinski, and Wu, 2022). Returns and alpha are presented daily in percentage terms. SR denotes the annualised Sharpe Ratio, with Newey-West (1987) t-statistics in parentheses. Bold figures indicate the highest Sharpe ratio for each ML model across the three datasets (OF, EF, OF+EF). The out-of-sample test periods range from February 18, 2020, to June 30, 2025. sets (O F, EF, OF+EF). The out-of-sample test sample periods are from February 18, 2020, to June 30, 2025.

Table 13: Momentum Factor Betas

$\beta_{momfactor}$			
	OF	EF	OF+EF
OLS (t-stat)	12.97 (-2.26)	-13.78 (-2.36)	-8.92 (1.56)
RidRegre	11.86	-14.80	-11.38

(t-stat)	(-2.08)	(-2.53)	(-2.00)
LASSo	-3.82	-7.34	-7.70
(t-stat)	(-0.76)	(-1.63)	(-1.71)
EN	-5.56	-11.18	-11.48
(t-stat)	(-1.07)	(-2.29)	(-2.35)
PCR	15.21	8.44	3.35
(t-stat)	(-2.75)	(-1.57)	(-0.65)
RForest	-11.30	6.17	-1.70
(t-stat)	(-2.20)	(-1.13)	(-0.31)
SGBD	-5.72	-1.44	-1.44
(t-stat)	(-1.11)	(-0.32)	(-0.32)
NN1	-3.76	-9.25	-14.91
(t-stat)	(-0.70)	(-1.56)	(-2.84)
NN2	-7.92	1.65	-0.49
(t-stat)	(-1.45)	(-0.28)	(-0.09)
NN3	-4.69	-19.07	-6.21
(t-stat)	(-0.87)	(-3.53)	(-1.11)
NN4	-16.08	-11.98	-17.45
(t-stat)	(-3.04)	(-2.39)	(-3.35)
L-Mean	16.17	-14.50	-10.13
(t-stat)	(-2.79)	(-2.45)	(-1.78)
NL-Mean	-9.10	-8.50	-7.55
(t-stat)	(-1.69)	(-1.54)	(-1.36)

Source: Author's Own Creation

Note: The table above presents the momentum factor betas for long-short portfolio returns, sorted by ML forecasts, relative to the three-factor model (Liu et al., 2022). The portfolios are equally weighted and recalculated daily. The daily results are provided for three information sets: OF, based solely on order flows; EF, based on economic fundamentals; and OF+EF, incorporating both. All models rely on lagged returns. The out-of-sample testing period runs from February 18, 2020, to June 30, 2025.

Table 14 Vector Autoregression Results

	BTC	ETH	USDT	BNB	XRP	USDC	SOL	TRX	DOGE	BCH	Top 10	All 128
Rate of Return Parity												
$OrFlow_{i,t}^W$	0.025***	0.035***	0.048***	0.032***	0.025***	0.050***	0.045***	0.026***	0.048***	0.048***	0.038***	0.023***
	(13.05)	(12.90)	(12.71)	(9.03)	(8.40)	(20.66)	(10.73)	(8.03)	(15.44)	(15.44)	(30.28)	(28.73)
$OrFlow_{i,t-1}^W$	0.004	0.000	0.008**	0.000	0.007**	0.008*	0.008*	-0.003	0.004	0.010***	0.005***	0.005***
	(1.26)	(0.27)	(20.22)	(0.14)	(2.32)	(2.20)	(1.82)	(-0.63)	(1.06)	(3.20)	(3.51)	(5.41)
$\sum_{j=1}^{j=5} OrFlow_{i,t-j}^W$	0.005	0.004	0.016**	0.012*	0.016***	0.018***	0.012	0.004	0.019***	0.019***	0.009***	0.006***
	(0.90)	(0.61)	(2.15)	(1.95)	(3.83)	(2.96)	(1.41)	(0.48)	(2.14)	(3.41)	(4.56)	(3.06)
$re_{i,t-1}$	-0.037	-0.018	-0.030	0.012	-0.068**	-0.048	0.046	0.003	-0.021	-0.060*	-0.018	0.000
	(-1.29)	(-0.63)	(-0.75)	(0.24)	(-2.49)	(-1.31)	(0.79)	(0.06)	(-0.72)	(-1.83)	(-1.17)	(-0.19)
$\sum_{j=1}^{j=5} re_{i,t-j}$	-0.59	0.014	-0.099	0.031	-0.136**	-0.164*	-0.026	0.026	-0.052	-0.131	-0.043*	-0.057*
	(-0.83)	(0.18)	(-0.97)	(0.41)	(-1.99)	(-1.74)	(-0.39)	(0.36)	(-0.76)	(-1.31)	(-1.70)	(-1.77)
R-squared (%)	28.5	35.3	44.0	21.3	16.2	47.6	39.3	13.5	43.3	37.0	31.5	7.8
Order Flow Parity												
$OrFlow_{i,t}^W$	0.062	-0.057	0.018	-0.153**	-0.027	-0.090*	0.041	-0.014	-0.026	-0.096	-0.035**	-0.013
	(1.54)	(-1.13)	(0.27)	(-2.56)	(-0.75)	(-1.81)	(0.65)	(-0.28)	(-0.59)	(-2.05)	(-2.05)	(-1.72)
$\sum_{j=1}^{j=5} OrFlow_{i,t-j}^W$	0.239***	0.008	0.061	-	-0.053	0.030	0.020	0.088	-0.040	-0.066	0.030	0.033***
	(3.05)	(0.09)	(0.45)	(-2.68)	(-0.76)	(0.25)	(0.25)	(1.11)	(-0.46)	(-0.69)	(0.69)	(2.45)
$re_{i,t-1}$	-	-0.005	-1.195*	0.444	-0.396	-0.605	-	-0.996	-1.315**	-0.889	-	0.436***
	2.333***						2.127***				0.949***	
	(-2.85)	(0.00)	(-1.85)	(0.72)	(-0.63)	(-0.85)	(-3.45)	(-1.84)	(-2.09)	(-1.39)	(-4.66)	(-5.95)
$\sum_{j=1}^{j=5} re_{i,t-j}$	-1.163	0.631	0.856	2.658*	0.236	0.015	-1.325	-0.396	0.299	0.005	-0.151	-0.335**
	(-0.67)	(0.49)	(0.65)	(1.90)	(0.17)	(0.02)	(-1.13)	(-0.35)	(0.26)	(0.02)	(-0.37)	(-2.09)
R-squared (%)	1.9	1.8	2.4	2.9	0.6	2.5	1.4	0.0	0.5	2.6	1.9	0.6

Table 15: Duple-types or double-sorted portfolios—economic fundamentals

Linear ML Forest Combination (L-mean)							
	Pr_1	Pr_2	Pr_3	$Pr_3 - Pr_1$			
	Mean	Returns	(% Daily)	Mean	Alpha	SR	Re_{oos}^2
Arbitrage 1	0.59	0.59	0.46	-0.13	-0.10	-0.64	-4.59
Index	(2.39)	(2.44)	(1.49)	(-0.97)	(-0.79)		
2	0.40	0.86	0.51	0.11	0.22	0.37	-2.32
	(1.50)	(3.28)	(1.94)	(0.56)	(1.18)		
3	0.23	0.62	1.03	0.79	0.92	2.53	-1.93
	(0.86)	(2.38)	(3.78)	(0.09)	(4.53)		
Non-Linear ML Forest Combination (NL-mean)							
	Pr_1	Pr_2	Pr_3	$Pr_3 - Pr_1$			
	Mean	Returns	(% Daily)	Mean	Alpha	SR	Re_{oos}^2
Arbitrage 1	0.50	0.61	0.54	0.03	0.07	0.20	0.31
Index	(2.12)	(2.53)	(2.15)	(0.31)	(0.48)		
2	0.36	0.63	0.79	0.43	0.49	1.62	0.32
	(1.41)	(2.46)	(2.86)	(2.55)	(2.78)		
3	0.19	0.79	0.91	0.72	0.82	2.34	0.12
	(0.75)	(2.69)	(3.44)	(3.45)	(4.04)		

Source: Author's Own Creation.

Note: The table presents the performance of the dual-sorted cryptocurrency portfolios for models subject to economic fundamentals (EF). The data is first sorted by the arbitrage index, using quantiles of size, idiosyncratic volatility, and illiquidity. The second sort is carried out by the linear ML forest combination (L-mean), which is based on EF, and the non-linear ML forecast combination (NL-Mean) for the EF model. The portfolios sorted are equally weighted and reconciled daily. The alpha, based on the three-factor model, includes the crypto market, size, and momentum factors [25]. The average returns and alphas are expressed in daily percentages. SR refers to the Sharpe Ratio. The Newey and West, *t-statistics* are shown in parentheses. The table also shows the Re_{oos}^2 in percentage form for both the ML models for each level of the arbitrage index. The out-of-sample test period is from February 18, 2020, to June 30, 2025.

Conclusion

The rapid rise of cryptocurrencies is creating a new branch of financial econometrics, focusing on out-of-sample predictions of crypto returns and the analysis of consistently profitable trading strategies. Our research enhances understanding of how economic indicators of order flow can forecast crypto returns and extends beyond these indicators. We provide empirical evidence of a non-linear relationship between order flows and crypto returns, explaining why non-linear machine learning models outperform linear ones based on order flows. Additionally, the study shows that this relationship varies across coins, indicating that not all are equally affected by the magnitude of the price push. When comparing the top crypto coins to the other 128, this relationship becomes stronger. The effectiveness of non-linear order flow methods remains robust despite economic constraints such as short-selling limits, transaction costs, and arbitrage restrictions, supporting the consistent impact of order flow on crypto returns. Our multivariate portfolio analysis demonstrates that non-linear order flow models generate profitable results not only for minute, illiquid, and highly volatile coins, but also for the top 20 coins by market cap, as further validated by objective economic measures.

This paper provides a comprehensive analysis of out-of-sample predictability in cryptocurrencies, using a large cross-sectional dataset of daily excess returns. It includes a wide

range of economic indicators across categories and employs recent machine learning techniques to forecast crypto returns. Combining these predictors with machine learning enhances the accuracy of out-of-sample forecasts for daily cross-sectional cryptocurrency returns, as measured by the out-of-sample mean squared deviation (MSD). These forecasts are then used to optimise portfolios, which offer significant utility to investors. Such investors allocate assets among individual cryptocurrencies, T-bills, and long-short portfolios (LSE) that include cryptocurrencies with the highest or lowest return forecasts. This strategy yields high Sharpe, Calmar, and Sortino ratios and generates notable alphas, consistent with the [25] cryptocurrency three-factor model. Order flow plays a key role in explaining and predicting cross-sectional cryptocurrency returns. In in-sample regression analysis, world order flow shows persistent correlation and a causal link with crypto returns. Using nonlinear machine learning techniques to analyse data from all global order flows suggests high out-of-sample forecasting accuracy. This is supported by the very high economic value, with a forecast Sharpe ratio of 0.86 for the long portfolio and 3.97 for the long-short portfolios. Overall, our results remain robust even when accounting for factors such as short-selling constraints, transaction costs, and arbitrage limits, providing strong empirical support for the influence of order flow on crypto returns.

In conclusion, the evidence indicates that order flow information is significantly relevant for understanding cryptocurrency returns. The economic value of ML models that do not condition on order flow is clearly explained by arbitrage limits evident in the stock, options, and crypto markets. Ultimately, observational evidence strongly indicates that order-flow information is important for predicting cryptocurrency returns and is robust to economic restrictions.

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